

Their hybrid dual-interface EEPROM is used for data logging.

Keeping safety in mind

When developing health monitoring solutions, one should remember that it is more often than not used by people with special health constraints or conditions. Hence the solution itself must be non-intrusive and harmless without adding to their strain or causing new complications.

The non-intrusive, cloud-based pregnancy monitoring solution offered by Wipro Healthcare is an interesting example. The solution, piloted in hospitals in Bengaluru and Delhi, has been deployed in collaboration with a UK-based start-up, which provides the device for measuring foetal heart rate, maternal heart rate and other vital signs. Wipro provides the embedded systems solutions along with storage and analytics technologies on mobile and cloud platforms.

The technology, based on the principle of ECG, is so sophisticated that it can detect and interpret even the faintest heartbeat of the foetus. The wearable device is small in size as against an ultrasound device, which is large and uncomfortable, especially for women in the last trimester. The technology is also non-intrusive for the foetus and there are no transmissions to the womb as like in the case of ultrasound devices. It accurately records maternal and foetal heart rate, and uterine activity, thereby providing information on foetal and maternal well-being. The device is suited for both antenatal care, and during active labour and delivery. The comprehensive solution helps gynaecologists to monitor patients 24/7 using information delivered to their mobile.

This as well as Wipro's cardiac care solution are based on the Wipro AssuranceHealth platform, which leverages Microsoft's cloud, mobility and analytics offerings to allow care providers to monitor patients regularly and precisely. This is done through hosted services and mobile apps that integrate medical devices, IT Infrastructure and

Personal fitness monitors

There are several personal health monitoring tools popping up in the market these days. Here are some quick examples:

Jawbone UP. It is a wristband that tracks the wearer's sleep time, diet and activities—24x7. It alerts you when you have been idle too long or when it is time for your routine exercises. You can discover patterns in your activities, analyse your fitness and more. The wristband syncs and charges using USB, and works with an iOS app. It includes a precision motion sensor, single push-button interface, vibrating motor for notifications and dual LEDs.

Larklife. This pair of wristbands too tracks the user's sleep, activity and diet. An intelligent triaxial accelerometer helps pick up even micro-movements of the body, while dietary information needs to be fed into the system. A continuous machine learning algorithm analyses your activity and learns about your energy patterns. The techniques are apparently based on the science of circadian rhythms, or how individuals' energy levels peak and dip. Larklife connects to an iOS app through Bluetooth Smart.

Addas miCoach. This chest strap instantly turns your Bluetooth Smart compatible phone into a personal trainer. It monitors your heart rate and helps you stay in the right zone as you run. It then wirelessly uploads your statistics to miCoach.com, where you can see the results, set goals, create training plans and get professional feedback. The app also tracks speed, distance and pace.

Others. Other such products include Fitbit, Nike Fuelband, Amigo, Basis and Misfit Wearable's Shine.

24x7 customer support, to deliver highly-scalable solutions.

Managing diversity and individual choices

When we speak of healthcare in a country, there is a lot of diversity. There is diversity in needs, culture, infrastructure and usage. So a telemedicine platform has to be able to encompass all of these smoothly.

"Cisco HealthPresence 2.5 gives a lot of choices to our customers. Our vision is to allow healthcare collaboration across any standard video endpoint, and Cisco provides almost ten choices from low-end desktop based to mid-end Tandberg-based to high-end truly-immersive-based video endpoints. In addition to high-quality audio and video, we believe in integrating with all the relevant medical devices and giving lots of choices to our customers based on standards-based medical device integration. We also want to make it very efficient for the doctors to do the consultation. So we have enabled integration with any EMR the customer may have, as well as integration with any kind of imaging modalities," says Gupta.

The Cisco healthcare solution leverages the network as a platform, and combines advanced audio, video and third-party medical devices to create

a high-quality clinical consultation environment enabling doctor-patient consultation remotely. Primary healthcare centres or community healthcare centres in rural areas are connected to the district hospital or a super-specialty hospital that hosts the healthcare solution. The process starts with a patient visiting a village healthcare centre where the point-of-diagnosis (PoD) is installed. Thereafter, the local paramedic initiates a call with the specialist, who consults as if they were face-to-face, and gives a prescription.

"While the platform can perform well from about 256 kbps, to get the optimal user experience, we recommend about 1 Mbps. The whole interaction is managed out of a cloud with some specialised software at the edge," adds Gupta.

The medical solution has been implemented in seven districts (Raichur, Shimoga, Chitradurga, Gwalior, Sehore, Datia and Chhindwara) in partnership with the state governments of Karnataka and Madhya Pradesh. As of December 2012, they had conducted 27,000 specialty and super-specialty consultations in just one year.

"Our aim is to provide healthcare solutions to the rural population in a manner that eliminates the need to travel long distances to obtain basic healthcare facilities. Our solutions are